

Numerical Analysis and Scientific Computing

Foundations, algorithms, and the software layer

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May 29, 2026

Abstract

We will cover numerical analysis from approximation and stability to large-scale algorithms, automatic differentiation, Monte Carlo, and probabilistic numerics.

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1 PART I — FOUNDATIONS

1.1 1. Errors, Conditioning, and Floating Point

- IEEE arithmetic; machine epsilon; rounding models
- Condition number of a problem (not just a matrix)
- Forward and backward error analysis; Wilkinson's program
- Catastrophic cancellation; examples of unstable algorithms

1.2 2. Polynomial Approximation and Interpolation

- Lagrange and Newton interpolation; divided differences
- Chebyshev nodes; Runge's phenomenon; optimal point selection
- Spline interpolation: cubic splines, B-splines, tension splines
- Barycentric interpolation formula; numerical stability

1.3 3. Approximation Theory

- Best approximation in normed spaces; existence, uniqueness
- Chebyshev equioscillation theorem; Remez algorithm
- Orthogonal polynomials: Legendre, Chebyshev, Hermite, Laguerre
- Rational approximation; Padé approximants

1.4 4. Numerical Differentiation and Integration

- Finite differences; Richardson extrapolation; accuracy
- Newton-Cotes formulas; error analysis
- Gaussian quadrature: nodes, weights, optimality
- Adaptive quadrature; Gauss-Kronrod; oscillatory integrals

2 PART II — CORE ALGORITHMS

2.1 5. Direct Methods for Linear Systems

- LU factorization; pivoting; stability analysis
- Cholesky for SPD; LDLT decomposition; tridiagonal systems
- QR factorization: Householder, Givens, modified Gram-Schmidt
- Least squares; pseudoinverse; rank-deficient problems; regularization

2.2 6. Iterative Methods for Linear Systems

- CG (SPD), MINRES (symmetric), GMRES (general), BiCGSTAB
- Convergence theory; spectral analysis; polynomial approximation view
- Preconditioning: ILU, incomplete Cholesky, algebraic multigrid (conceptual)
- Choosing a solver; practical considerations

2.3 7. Eigenvalue Algorithms

- Power method; inverse iteration; shift-and-invert
- QR algorithm with shifts and deflation; convergence theory
- Symmetric QR with Wilkinson shift; divide-and-conquer
- Krylov methods: Lanczos (symmetric), Arnoldi (general); thick restart

2.4 8. Nonlinear Equations and Fixed Points

- Bisection; secant method; Newton's method; convergence analysis
- Newton's method in $\mathbb{R}^n$; convergence, local vs. global
- Fixed-point iterations; Banach contraction; Aitken-Steffensen acceleration
- Homotopy and path-following methods; applications

2.5 9. Numerical ODEs

- One-step methods: Euler, Runge-Kutta, order conditions
- Multi-step methods: Adams, BDF; zero-stability; Dahlquist barrier
- Stiff systems; implicit methods; A-stability; L-stability
- Event detection; discontinuities; delay equations

2.6 10. Numerical PDEs

- Finite differences: elliptic, parabolic, hyperbolic; stability analysis
- Finite element methods: weak form, assembly, error analysis
- Spectral methods: Fourier, Chebyshev, pseudospectral collocation
- Multigrid: V-cycle, smoothers, convergence theory

3 PART III — MODERN METHODS

3.1 11. Optimization Algorithms

- Gradient descent and line search; Newton and quasi-Newton (BFGS, L-BFGS)
- Interior point methods; barrier approach
- Nonlinear least squares: Gauss-Newton, Levenberg-Marquardt
- Derivative-free optimization: Nelder-Mead, pattern search (brief)

3.2 12. Automatic Differentiation

- Forward mode via dual numbers; mathematical formalization
- Reverse mode (backpropagation) via computational graphs; chain rule at scale
- Jacobian-vector products (JVPs) and vector-Jacobian products (VJPs)
- Higher-order AD: forward-over-reverse; Hessian-vector products
- JAX, PyTorch autograd, Zygote (Julia) as implementations
- Adjoint method for PDE-constrained optimization; connection to Course 3 §22

3.3 13. Monte Carlo and Quasi-Monte Carlo Methods

- Classical Monte Carlo; variance estimation; confidence intervals
- Variance reduction: importance sampling, control variates, stratification
- Markov chain Monte Carlo: Metropolis-Hastings, Gibbs sampler
- Hamiltonian Monte Carlo; MALA
- Quasi-Monte Carlo: Halton and Sobol sequences; Koksma-Hlawka inequality
- Applications: Bayesian computation, uncertainty quantification, rare events

3.4 14. High-Dimensional Numerical Methods

- Curse of dimensionality: what breaks and what survives
- Sparse grids: Smolyak construction; dimension-adaptive variants
- Tensor decompositions in numerical PDE: TT/MPS, Tucker
- Sparse polynomial chaos; stochastic collocation for UQ
- Importance for quantum chemistry, financial mathematics, statistical learning

3.5 15. Randomized Linear Algebra

- Randomized SVD (Halko-Martinsson-Tropp); single-pass algorithms
- Johnson-Lindenstrauss lemma; random projections; sketching
- Randomized least squares; sketch preconditioners
- Nyström approximation for kernel matrices; applications

3.6 16. Structured Matrices and Fast Algorithms

- Toeplitz and circulant matrices; FFT as diagonalization
- Hierarchical matrices (H-matrices): structure, arithmetic
- Fast multipole method (conceptual)
- Applications: structured covariance, integral equations

3.7 17. Matrix Functions

- Matrix exponential: scaling-squaring with Padé; Krylov methods for large systems
- Other functions: square root, logarithm, sign, cosine
- Fréchet derivative of matrix functions; algorithmic differentiation
- Applications: network centrality, Markov chains, differential equations

3.8 18. Probabilistic Numerics

- Bayesian quadrature: integration with uncertainty quantification
- Gaussian process models for ODE solutions; probabilistic linear solvers
- Connections between numerical algorithms and Bayesian inference
- Uncertainty-aware scientific computing